

Federal Urdu University

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Section: B

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Movie & Show Recommendation Agent

1. Agent Explanation

The **Movie & Show Recommendation Agent** is a lightweight software agent that suggests movies/shows to a user based on explicit preferences (type, format, genre).

Purpose: To provide fast, transparent, and predictable recommendations without requiring user history, ratings data, or internet connectivity.

How it works: The agent applies **if-then rules** (a decision tree) to match user-selected criteria (e.g., Type = Movie, Genre = Comedy) against a **static local dataset** (dictionary/database) of pre-tagged titles.

Why rule-based, not ML/API-based:

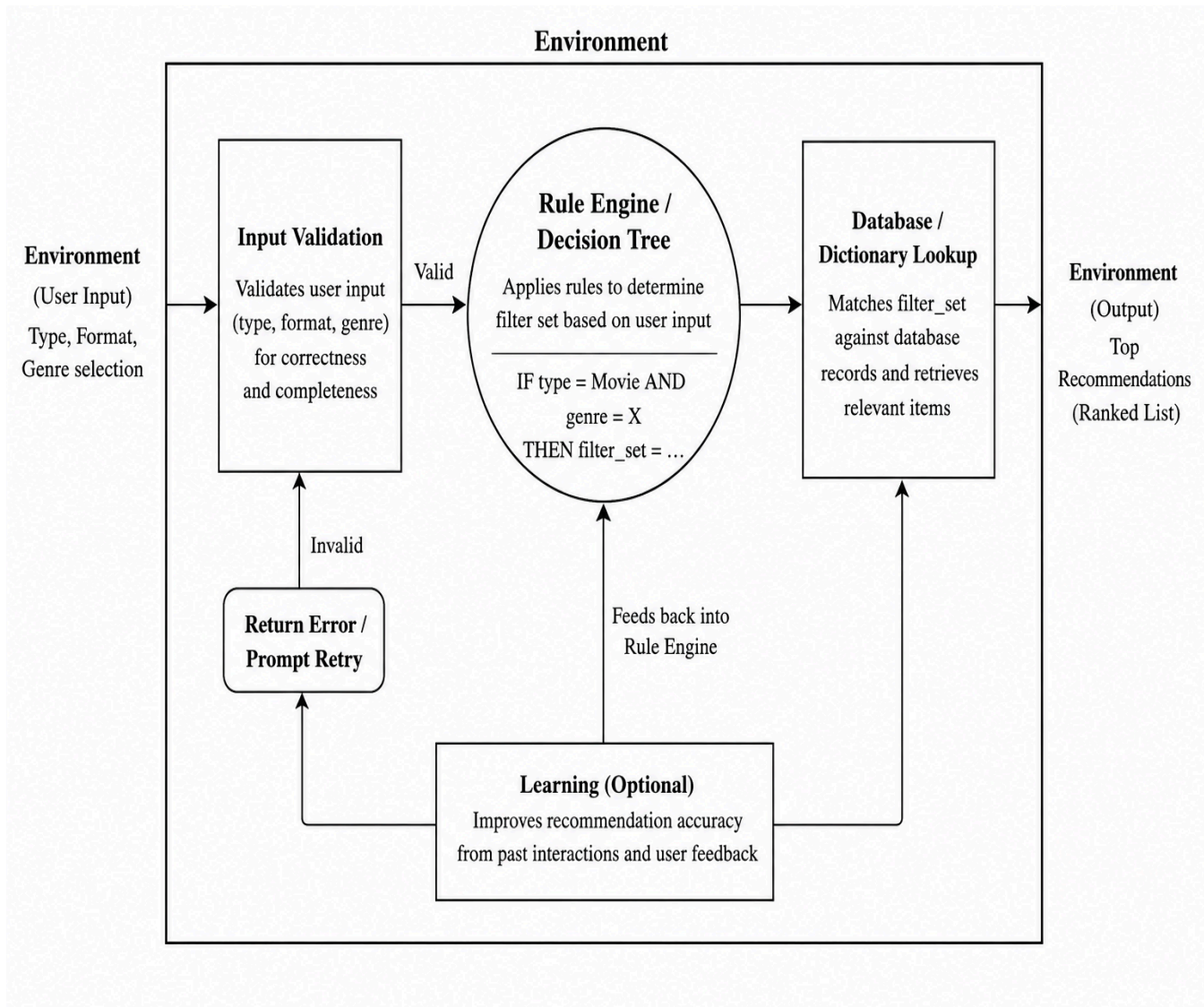
No training data or model tuning required, logic is human-readable and easy to debug.

Fully deterministic and explainable, every recommendation can be traced to a specific rule.

No external API dependency, works offline, with zero latency, cost, or privacy concerns.

Suitable for small, well-defined datasets where pattern complexity doesn't justify ML overhead.

2. Graphical Representation (Execution Flow)



3. Agent Formulation

Property	Description
Agent Type	Simple Reflex Agent (with rule-based/goal-based decision layer)
Fully/Partially Observable	Fully Observable — all inputs (type, genre, format) are directly visible to the agent
Deterministic/Stochastic	Deterministic — same input always produces the same output
Static/Dynamic	Static — the dataset does not change during a single decision cycle
Discrete/Continuous	Discrete — finite set of genres, types, and formats
Single/Multi-Agent	Single-Agent — no interaction with other agents
Sensors	Form UI inputs, dropdown selections, text fields
Actuators	Screen display, output UI panel, recommendation list viewer

4. PEAS Framework Evaluation

PEAS Component	Details
Performance Measure	Recommendation relevance, response speed, user satisfaction, correctness of filtering
Environment	User interface (web/app form), local movie/show database or dictionary
Actuators	Display screen, recommendation cards/list, output panel
Sensors	User preference selections (dropdowns, buttons), click/input events